David Liu

ML Research Engineer & Senior Data Scientist · 14 years writing code, a decade of production ML across biotech, e-commerce, healthcare, recruiting, and fintech.

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SUMMARY

I build production machine learning systems end-to-end — from data ingestion and feature engineering through model training, serving, and measurement — and I instrument them to verify their own claims. My work has a consistent pattern: question the accepted metric, find the architectural gap, build the system others extend into production. Currently leading personalization at Shipt for millions of grocery shoppers; previously a Machine Learning Research Engineer at Freenome (multiomics cancer detection) and a Sr. ML Engineer at Change Healthcare. I also run an independent AI research program — three papers spanning multi-agent synthesis, model selection under distribution shift, and LLM-grounded feature extraction — with evaluation designed so the system that produces a number is never the one that grades it.

EXPERIENCE

Senior Data Scientist — Shipt (Personalization Team)

SEP 2024 — PRESENT

Lead data scientist on Shipt's Personalization team; architected recommender and eligibility systems behind personalized shelves for millions of grocery shoppers. Mentored 4 data scientists, including 2 direct reports.

- Built and launched Deals For You V2, replacing matrix factorization with two-tower retrieval and precision reranking; a CUPED-corrected A/B test drove **+91% shelf engagement**, **+1.76% order incidence**, and **+1.41% orders per user**.
- Audited pre-treatment imbalance and a variant activation gap before SVP planning, correcting the annualized estimate from +83K to **~50K incremental orders (95% CI: 20K–80K)** and **~$3.7M directional GMV**; preserved a real win while retiring an inflated topline.
- The validated result prompted leadership to re-plan H2 around Deals as a potential **~$6M contributor to Shipt's $20M incremental-GMV target**, prioritizing scale-up, holdback measurement, and reuse of the eligibility platform.
- Architected a reusable real-time Eligibility platform: a store-scoped hard pre-filter before ANN with swappable content tables; reduced an initial **448M-row design by ~95%** to ~20M store-specific rows plus a compact global set, with zero-downtime refreshes and p99 <750 ms serving.
- Across the broader personalization portfolio, delivered up to **23% Personalized GMV lift** across A/B-tested shelves and a cold-start strategy with **+47% engagement** and **+8% first-time orders**.
- Created the Personalization Interaction Score (views → clicks → ATC → purchases) and an automated Shelf Attribution pipeline; challenged an irreproducible 5% attribution claim, established the defensible 1–4% range, and raised the organization's measurement standard.
- Partnered with Engineering to replace CSV-based recommendation delivery with Kafka pipelines across legacy recommenders; served as the team's sole data scientist for its first four months while maintaining 16 repositories.
- Designed a Retrieval-Augmented Generation system over Shipt's retailer catalogs and built the business case that secured investment in an agentic-AI framework.

Machine Learning Research Engineer — Freenome
NOV 2020 — JUN 2024

Core ML engineer at a genomics company developing a blood test for early-stage cancer detection. Worked at the intersection of infrastructure and research.

- Key contributor to Freenome's core multiomics cancer-detection model that predicts cancer stage (1–4) from blood-draw data; built data abstractions for petabyte-scale genomic datasets that unblocked cross-analyte feature development and accelerated training/evaluation cycles.
- Built a model-comparison system tracking research-vs-production model performance side-by-side — required for FDA audit compliance, gave the team confidence that production models stayed aligned with research intent.
- Designed and built large portions of Freenome's distributed ML training and serving platform used daily by 30+ scientists; scaled CPU-bound training across O(100) machines, supported leave-one-out / K-fold evaluation, and built reproducible model-artifact storage.
- Led adoption of PyTorch, MLFlow, and RayTune across the ML team — replaced legacy tooling to unlock GPU acceleration, experiment tracking, and hyperparameter tuning at scale.
- Built a cloud-cost monitoring system surfacing the biggest GCP storage and compute expenses; visibility alone drove optimizations saving the company **over $10M annually** — one of the highest-ROI projects I've worked on.
- Recognized with a Servant Leadership Award, elected by managers and peers across the engineering organization.

Sr. Machine Learning Engineer — Change Healthcare
JAN 2020 — NOV 2020

ML systems for health-insurance claims processing — accuracy translates directly into operational cost savings.

- Designed a ranking model matching human workers to claims tasks based on skill, history, and complexity; **$7M annual value** by reducing manual task assignment and the need for additional hires.
- Built a classification model partitioning sensitive patient documents (image + text) to route claims to the correct processing workflow.
- Developed internal AWS tooling and production API infrastructure for the ML team's deployment pipeline.
- Led a cross-functional tiger team prototyping a conversational chatbot (Rasa + HuggingFace NLP) for internal claims-inquiry workflows.

Data Scientist — Riviera Partners
JAN 2019 — DEC 2019

ML models for an executive-recruiting firm; full pipeline from data collection to model serving.

- Developed a model suite: a job-departure-likelihood classifier, a regression model predicting team sizes from resume features, and a candidate-ranking model using a custom NDCG listwise loss.
- Built an end-to-end framework for rapid model prototyping, training, evaluation, and serving — enabled the team to iterate on new models without re-engineering infrastructure each time.
- Wrote scrapers harvesting structured candidate data from public sites and APIs.

Undergraduate Researcher — UC Berkeley
JAN 2017 — DEC 2018

- **California Institute for Energy and Environment (CIEE)** — built a recurrent neural network for predicting building energy usage, exploring how temporal consumption patterns can inform smarter grid management.
- **Bengson Research Lab, Sonoma State** — applied ML to EEG data to predict individualized occipital lobe activation patterns; demonstrated early feasibility for brain-computer interface applications.

Earlier Roles
2015 — 2017

- **Data Science Intern, Castlight Health** (2017) — entity matching and deduplication pipeline using gradient-boosted classifiers with hard-negative mining; 85–95% precision/recall across hospital, facility, and practitioner entities.
- **Data Science Contractor, Riviera Partners** (2016) — team-size prediction model from public data; Python wrapper for survival-model time-series analysis; Flask model-serving infrastructure.
- **URAP, Berkeley Institute of Data Science** (2016) — mapped UC Berkeley course progression across majors via class-taxonomy organization and deduplication.
- **Data Science Intern, Doximity** (2015) — gradient-boosted classifier for malformed scraped articles; reverse geocoding + fuzzy string matching to link doctors in news articles to facility profiles.

SELECTED PROJECTS

Adaptive Domain Intelligence — LLM-as-Feature-Engineer Architecture
2026 · RESEARCH · WORKING PAPER

Research architecture for compressing qualitative records (court opinions, clinical notes) into calibrated predictions without trusting the LLM: the model only proposes feature schemas and performs verbatim-quote-grounded extraction — unsupported values are dropped and counted toward a measured hallucination rate — while a small calibrated classical model makes the prediction; self-improvement is gated by a domain expert and a frozen gold set (permutation test + subgroup-fairness veto). Validated as a known-answer test on four real legal corpora: re-derives six decades of established judicial-politics signals and nulls, and correctly lands at chance on an ECtHR leakage trap where prior text-only work reported ~79%. An automated verifier re-derives every headline number (8/8, zero drift).

Meta Council — Multi-Expert AI Decision Support Platform 2025 — PRESENT · RESEARCH & PRODUCT

Multi-agent LLM framework where N expert agents (each with a unique persona and analytical framework) analyze queries in parallel, then a weighted synthesis step produces structured decision documents with confidence scores, dissent preservation, and risk matrices. Evaluated across 750+ benchmark runs spanning 6 domains and 5 models (3B to frontier-class): synthesis outperforms single-best by 29–58% ($p<0.0001$, $d=2.16$); the optimal aggregation method is domain-dependent; synthesis amplifies model quality non-linearly. Published as an independent research paper.

AutoTrader — ML-Powered Stock Prediction System 2024 — PRESENT · PERSONAL

Fully autonomous market-prediction system: collects nightly market data for 600+ tickers, engineers 500+ features across 8 source families, trains 1,800+ dual models (classifier for direction, regressor for magnitude), and delivers confidence-ranked predictions before market open every weekday. Built every piece — data ingestion, custom feature store, dual-model training framework with walk-forward validation and Optuna, FAISS-powered similarity search, tiered email subscription system with Stripe billing, multi-cloud GCP+Azure infrastructure — running autonomously for ~$235/month.

Patterns of Choice — Revealed vs. Stated Ethical Values 2026 · RESEARCH · OPEN-SOURCE

Longitudinal instrument measuring the gap between the ethical values people claim and the values their choices reveal, built around a pre-registered hypothesis that recurring-character narrative immersion reduces social-desirability bias. Shipped as a local-first offline PWA — event-sourced engine, deterministic on-device scoring, nothing leaves the device — with a 61-scenario corpus, five recurring-character arcs, and a cross-implementation parity gate between the app runtime and the analysis pipeline.

XGBoost Visual Guide 2026 · OPEN-SOURCE

Interactive visual textbook explaining XGBoost and gradient boosting from first principles — 10 sections covering decision trees, ensemble methods, animated step-by-step gradient boosting, learning rate effects, XGBoost-specific innovations (histogram splits, sparsity handling), early stopping, feature importance, and a hyperparameter cheat sheet. D3.js + Chart.js.

PUBLICATIONS

Adaptive Domain Intelligence: An LLM-as-Feature-Engineer Architecture for Compressing Qualitative Records into Calibrated Empirical Predictions

2026 · Independent Research · Working Paper · LLM feature engineering, grounded extraction, hallucination measurement, self-improving systems, calibration, legal NLP

Weighted Multi-Expert Synthesis for High-Stakes Decision Support: A Multi-Agent LLM Framework with Dissent Preservation

2026 · Independent Research · CC BY 4.0 · Multi-agent systems, LLM, decision support, weighted synthesis, dissent preservation, confidence calibration

Stability Bonus Regularization for Model Selection Under Positive-Class Distribution Shift

2026 · Independent Research · CC BY 4.0 · Model selection, distribution shift, cross-validation, class imbalance, regularization

EDUCATION

University of California, Berkeley — BS, Computer Science and Data Science (Dual Degree), Class of 2018. Berkeley Institute of Data Science Undergraduate Research Apprenticeship (2015).

TECHNICAL SKILLS

LANGUAGES	Python · SQL · Bash · Java · C/C++
ML & DATA	PyTorch · XGBoost · Scikit-learn · Pandas · MLFlow · FAISS · Gensim · NLTK · SpaCy
LLM & AGENTS	Claude & GPT APIs · multi-agent orchestration · RAG · grounded extraction · LLM evaluation & hallucination measurement · agentic coding workflows
EVALUATION & TESTING	evaluation harnesses & known-answer tests · benchmark design · A/B testing · walk-forward validation · permutation tests · calibration · pre-registered hypotheses
CLOUD	GCP · AWS · Azure
DATA INFRASTRUCTURE	PostgreSQL · Snowflake · MySQL · Spark · Kafka
ORCHESTRATION	Airflow · Flyte · Metaflow · GitHub Actions
INFRASTRUCTURE	Docker · Kubernetes · Git · CI/CD